

Chapter 1

Introduction

A note on the

We're making these

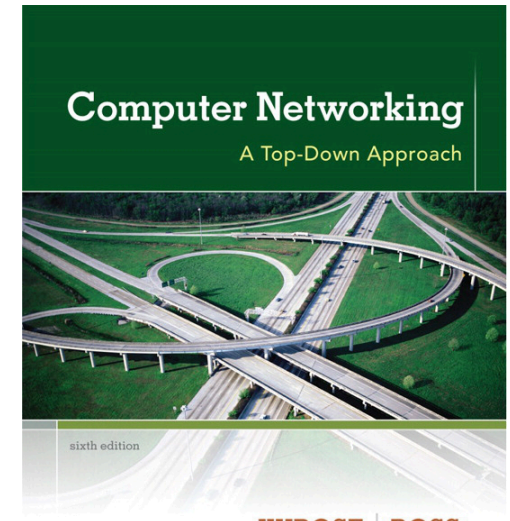
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Thanks and enjoy! JFK/KWR

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*Networking: A
Top Down
Approach
6th edition
Jim Kurose, Keith
Ross
Addison-Wesley
March 2012*

Chapter 1: introduction

our goal:

- ❖ get “feel” and terminology

overview:

- ❖ what’s the Internet?
- ❖ what’s a protocol?
edge; hosts, access

cour

- ❖ approach:
 - use Internet as example

uit
t structure

- ❖ performance: loss, delay, throughput
- ❖ security
- ❖ protocol layers, service models
- ❖ history

Chapter 1: roadmap

1.1 what *is* the Internet?

1.2 network edge

- end systems, links

structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

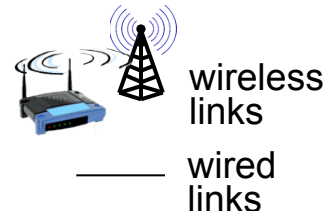
1.7 history

What's the Internet: "nuts and bolts" view



❖ millions of connected computing devices:

- *hosts* = *end syste*

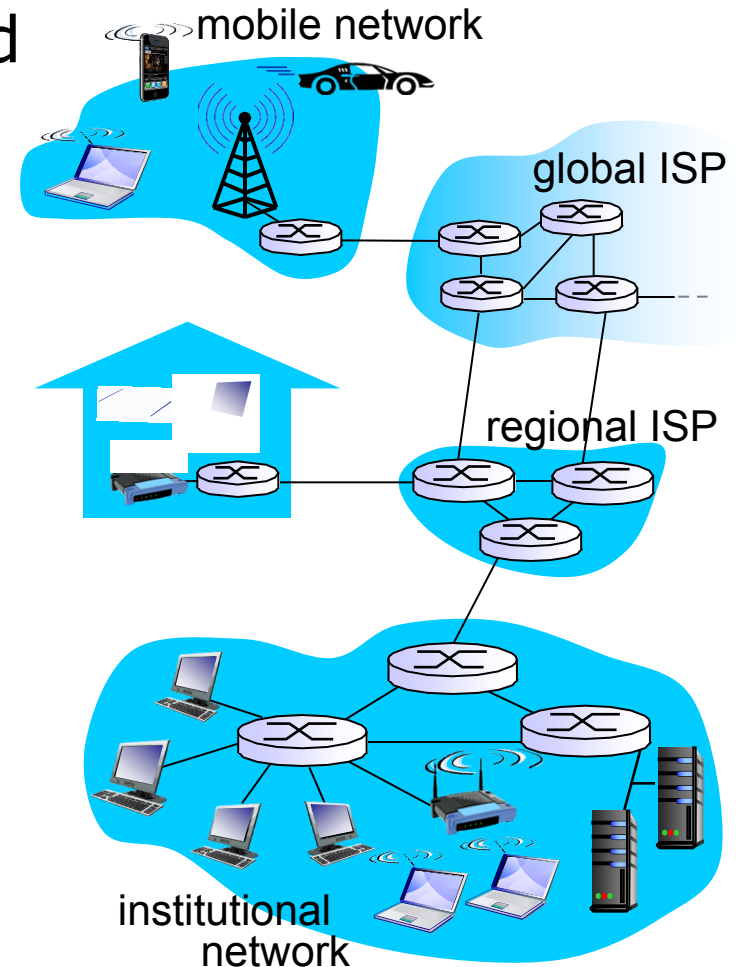


- fiber, copper, radio, satellite
- transmission rate:
bandwidth



❖ *Packet switches*:
forward packets
(chunks of data)

- *routers* and *switches*



“Fun” internet appliances



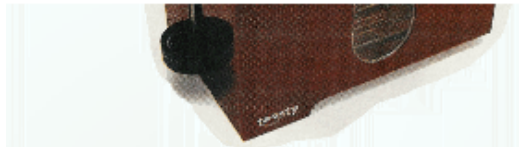
Web-enabled toaster +
weather forecaster



Tweet-a-watt:
monitor energy use



Internet
refrigerator



Slingbox: watch,
control cable TV remotely



Internet phones

What's the Internet: "nuts and bolts" view

❖ *Internet: "network of networks"*

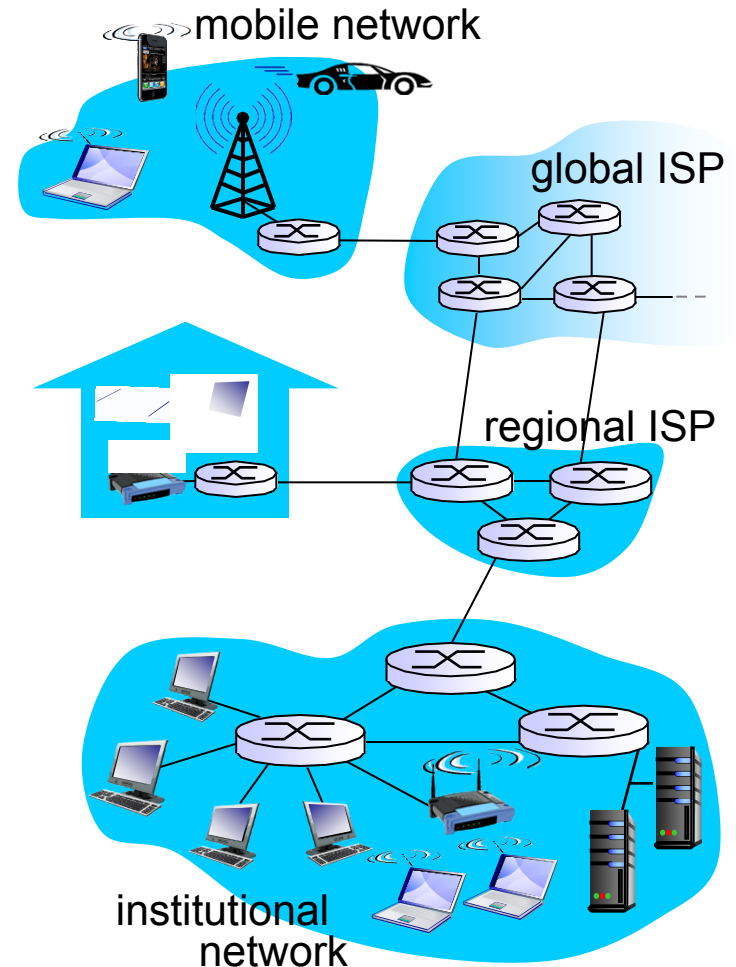
- Interconnected ISPs

❖ *protocols* contro

802.

❖ *Internet standards*

- RFC: Request for comments
- IETF: Internet Engineering Task Force



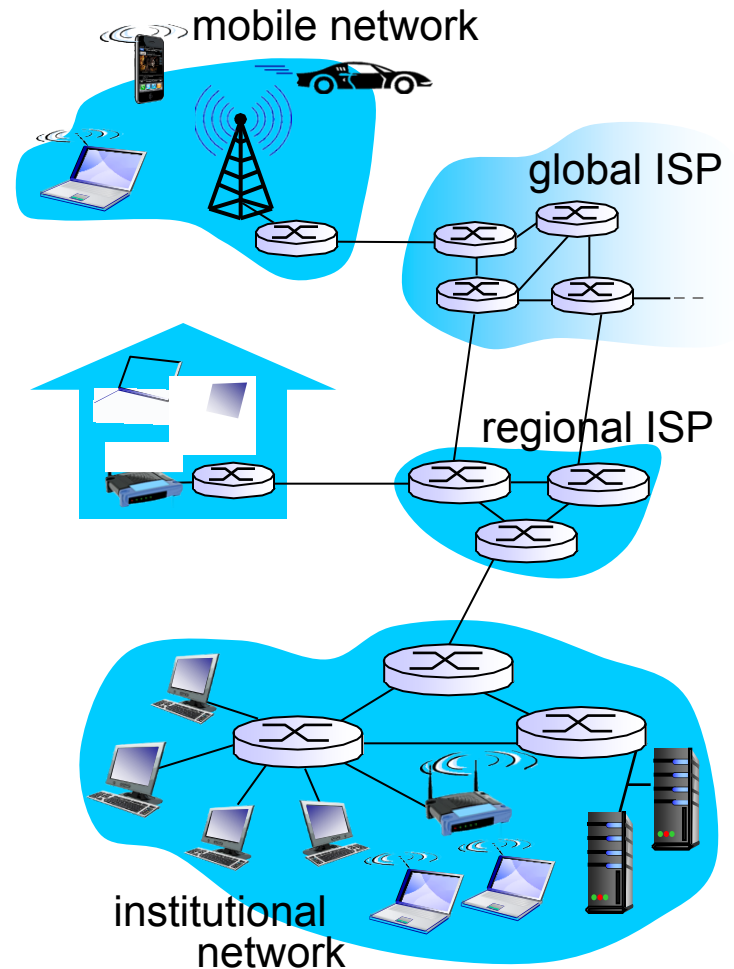
What's the Internet: a service view

❖ *Infrastructure that provides services to applications:*

soci

❖ *provides programming interface to apps*

- hooks that allow sending and receiving app programs to “connect” to Internet
- provides service options analogous to



What's a protocol?

human protocols:

- ❖ "what's the time?"
- ❖ "I have a question"
- ❖ introductions

network protocols:

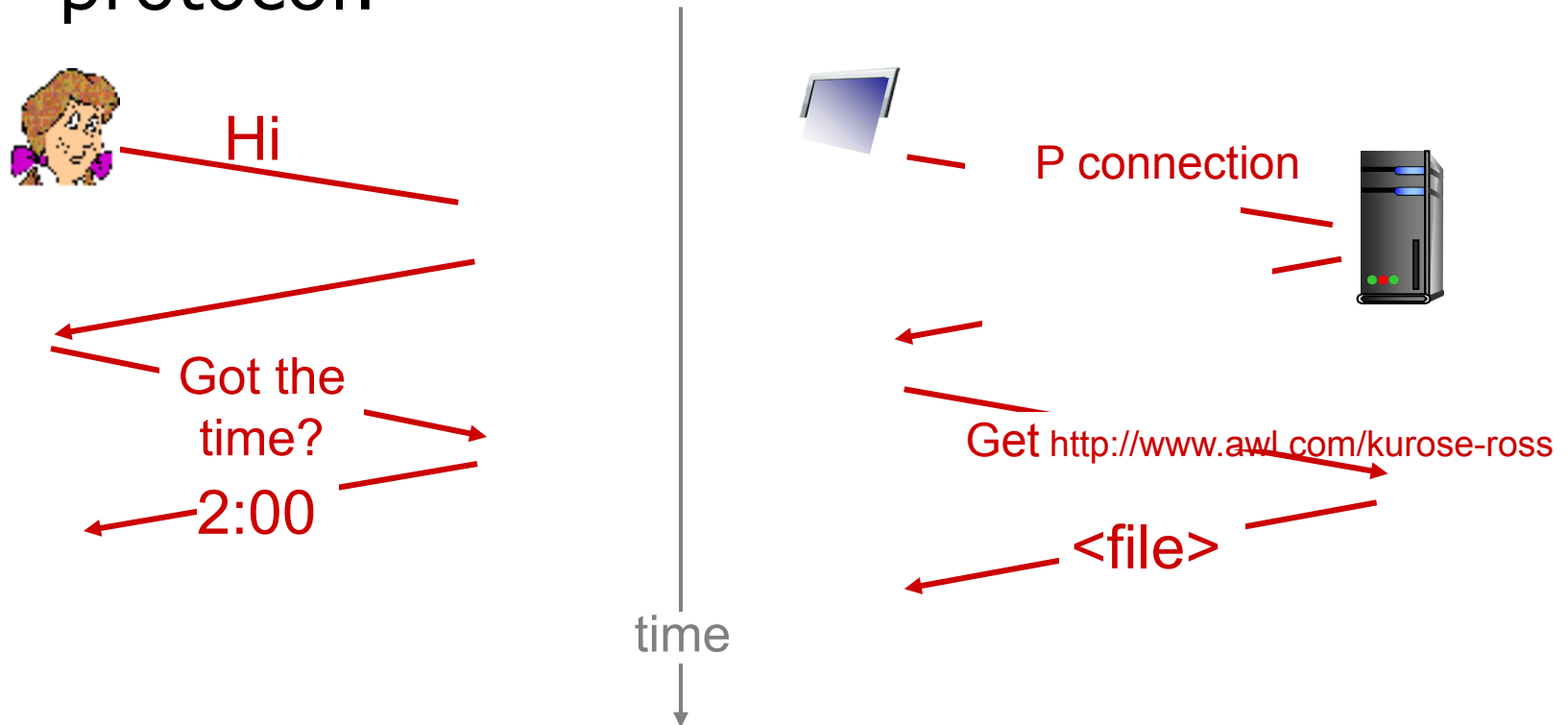
- ❖ machines rather than humans
communication

... specific actions taken when msgs received, or other events

protocols define format, order of msgs sent and received among network entities, and actions taken on msg

What's a protocol?

a human protocol and a computer network protocol:



Q: other human protocols?

Chapter 1: roadmap

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1.2 network edge

- end systems, links

network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

A closer look at network structure:

❖ *network edge:*

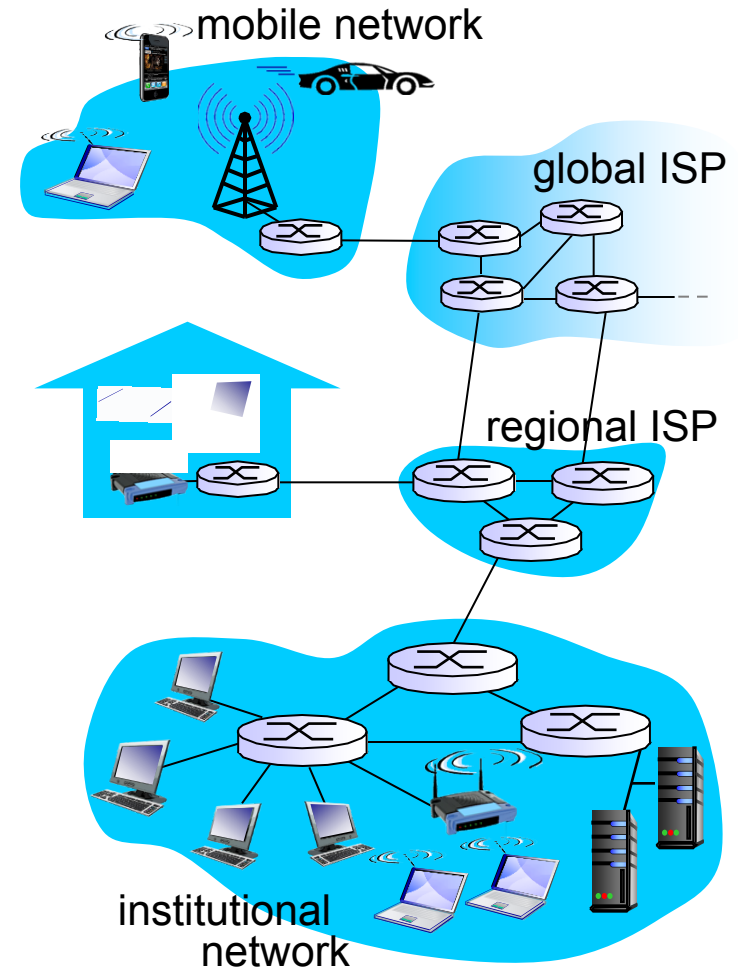
- hosts: clients and servers
- servers often



physi
wired, wireless
communication
links

❖ *network core:*

- interconnected routers
- network of networks



Access networks and physical media

Q: How to connect end systems to edge router?

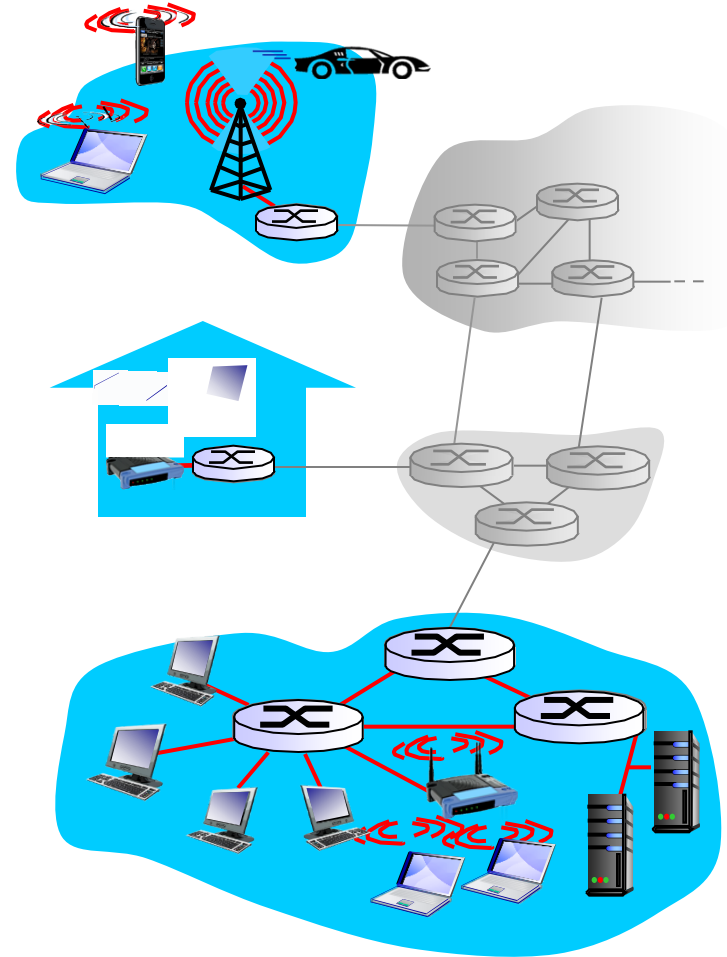
- ❖ residential acc

comp

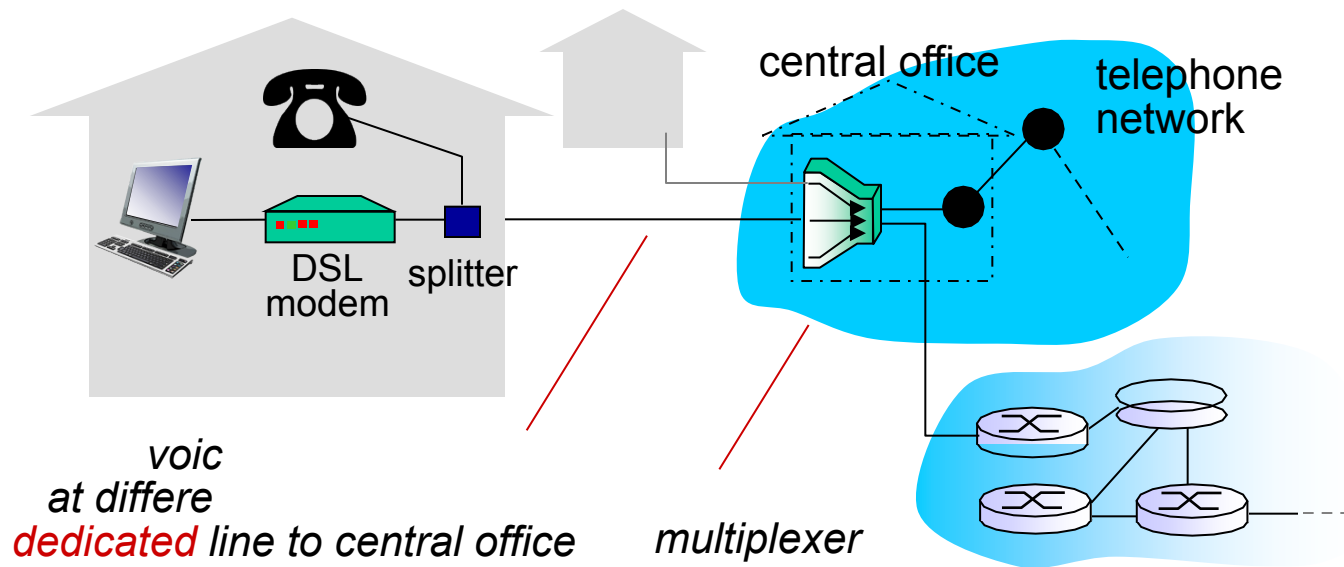
- ❖ mobile access networks

keep in mind:

- ❖ bandwidth (bits per second) of access network?
- ❖ shared or dedicated?

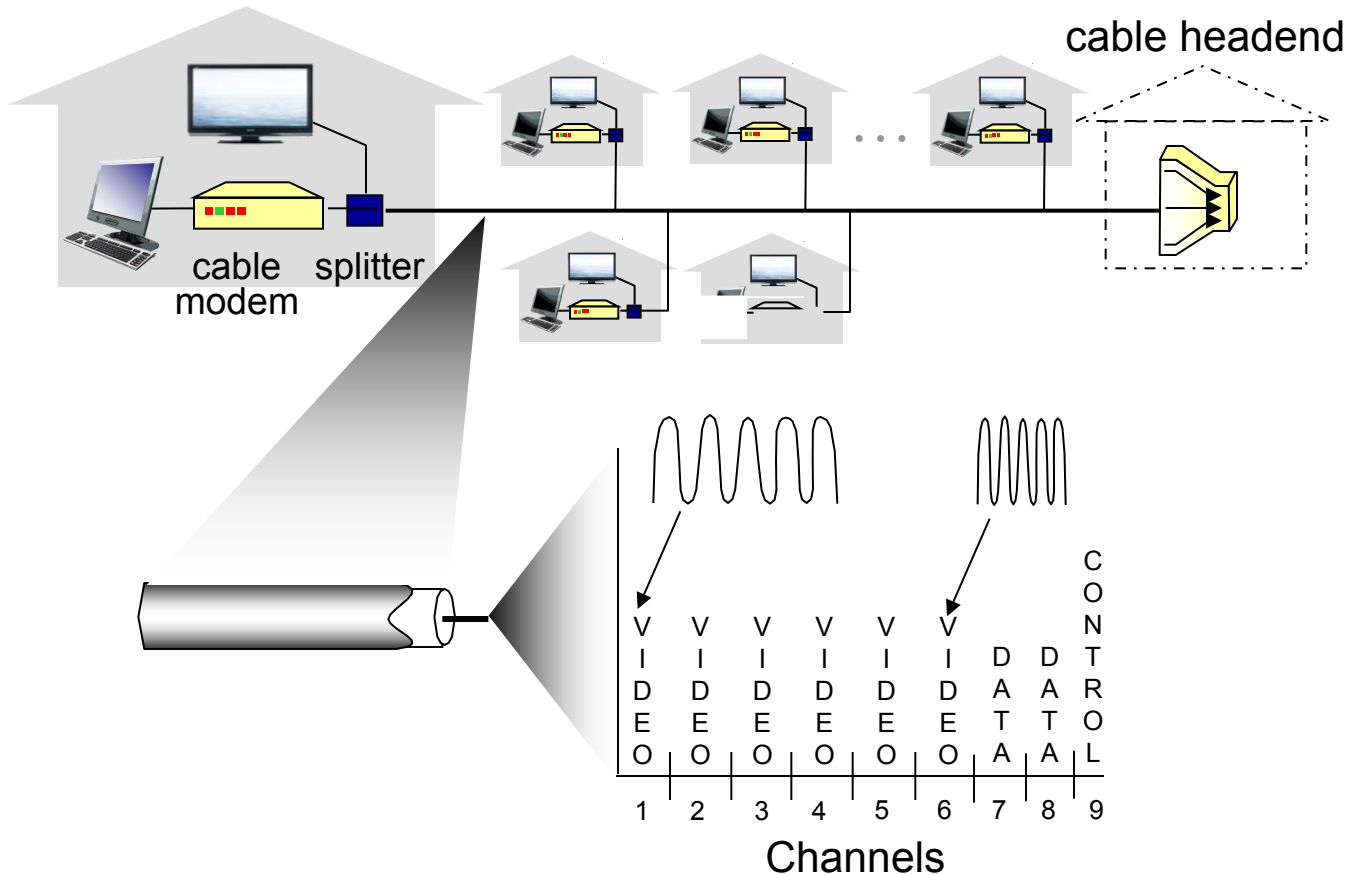


Access net: digital subscriber line (DSL)



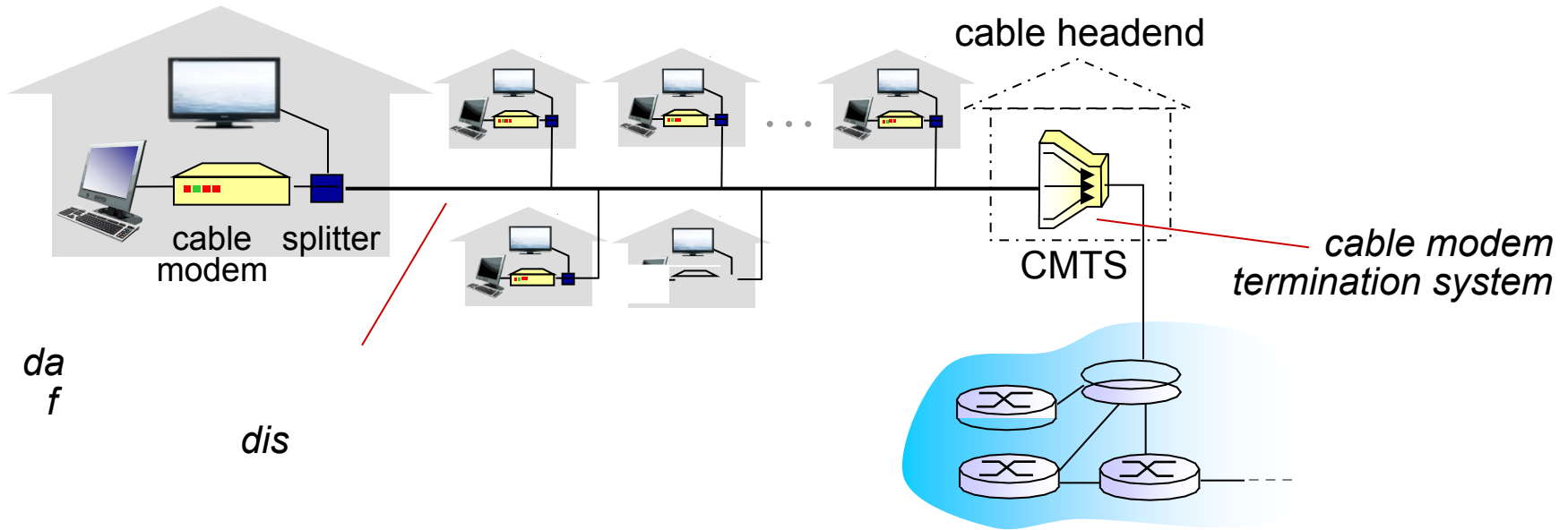
- ❖ use *existing* telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- ❖ < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- ❖ < 24 Mbps downstream transmission rate (typically < 10 Mbps)

Access net: cable network



frequency division multiplexing: different channels travel in different frequency bands

Access net: cable network



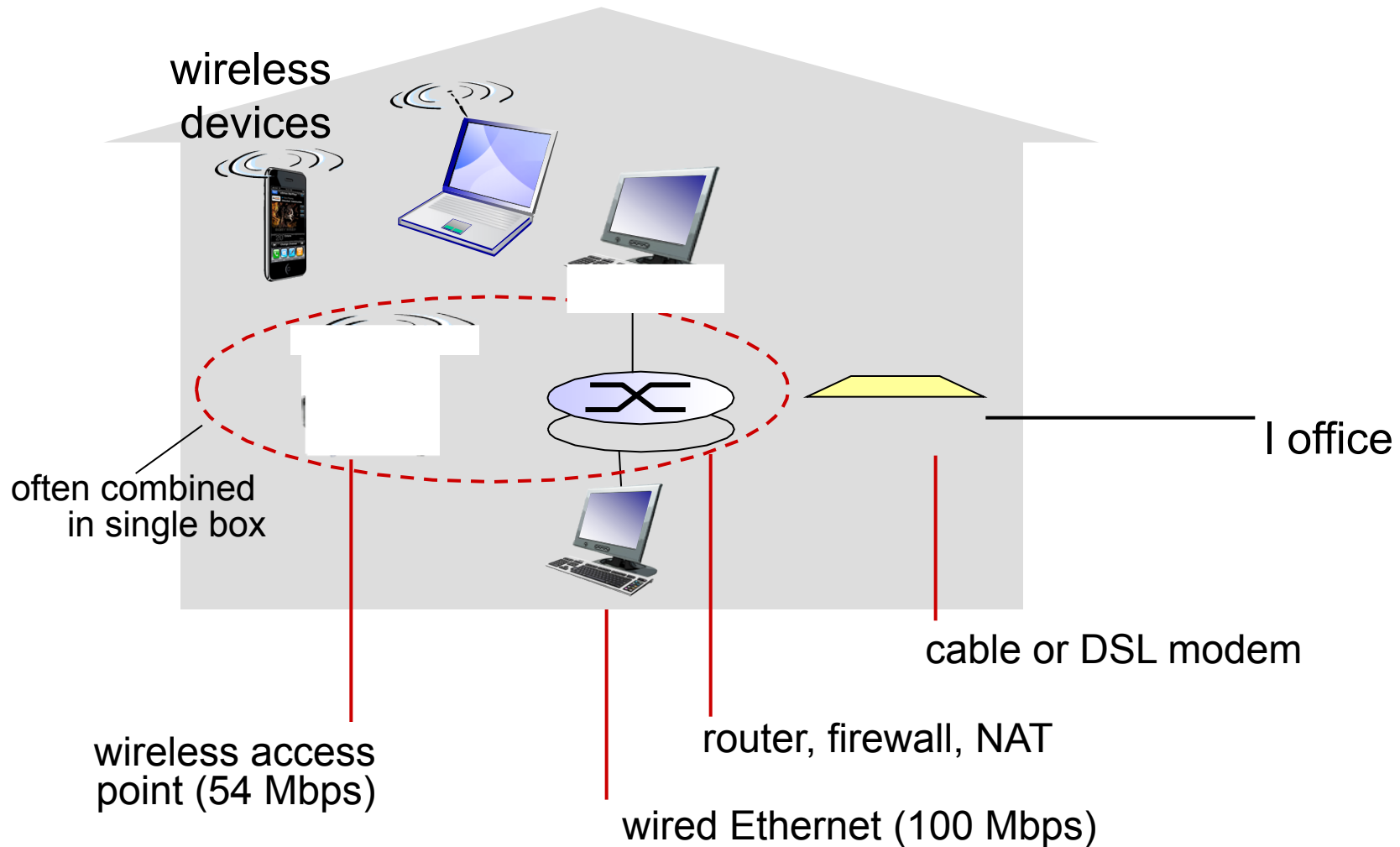
❖ HFC: hybrid fiber coax

- asymmetric: up to 30Mbps downstream transmission rate, 2 Mbps upstream transmission rate

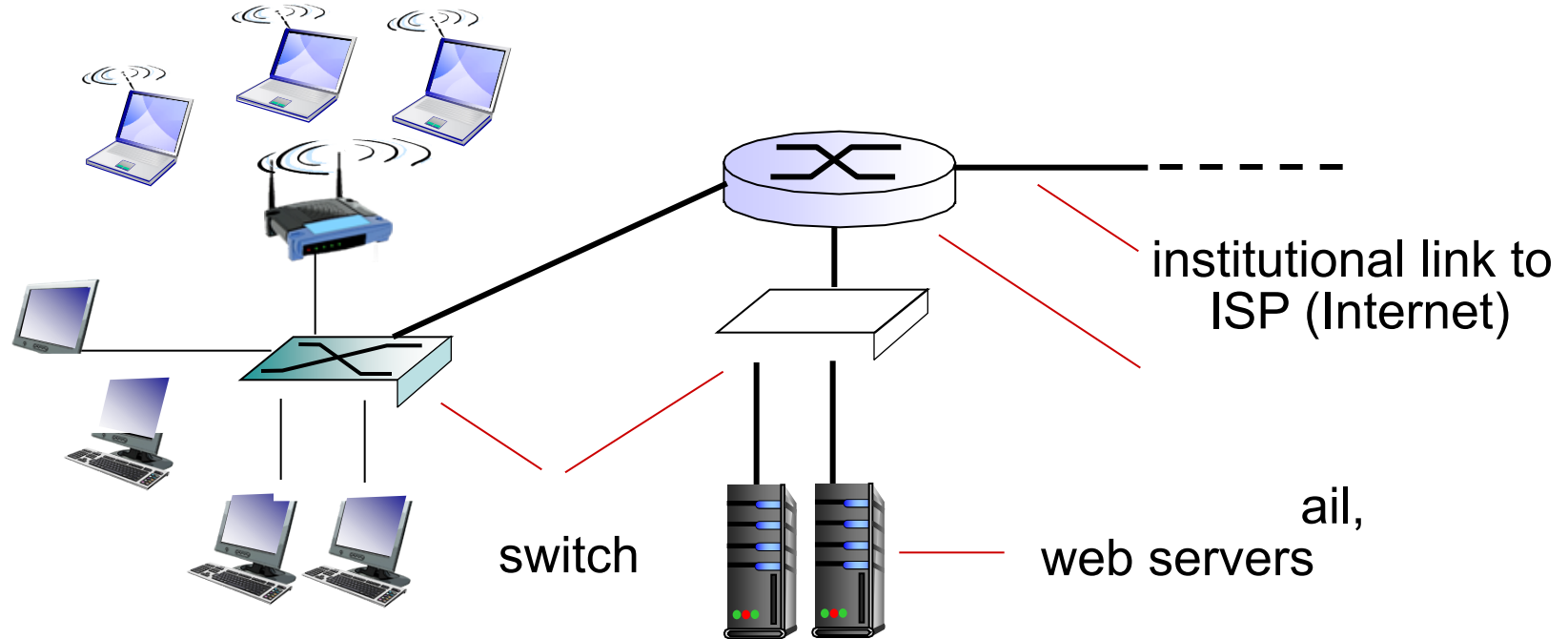
❖ network of cable, fiber attaches homes to ISP router

- homes *share access network* to cable headend
- unlike DSL, which has dedicated access to

Access net: home network



Enterprise access networks (Ethernet)



- ❖ typically used in companies, universities, etc
- ❖ 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- ❖ today, end systems typically connect into Ethernet switch

Wireless access networks

- ❖ shared *wireless* access network connects end system to router

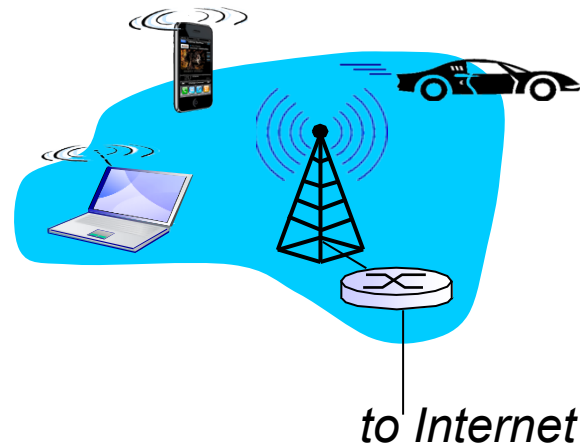
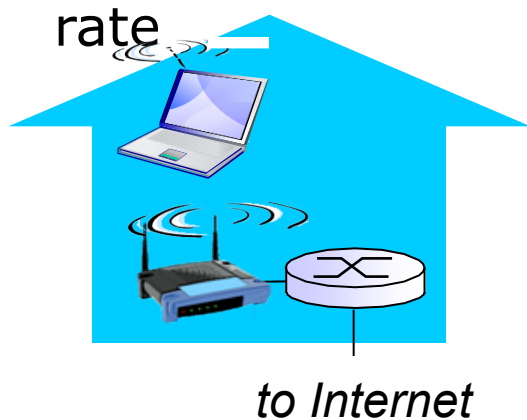
- via base station aka “access point”

wireless LANs:

- within building (

wide-area wireless access

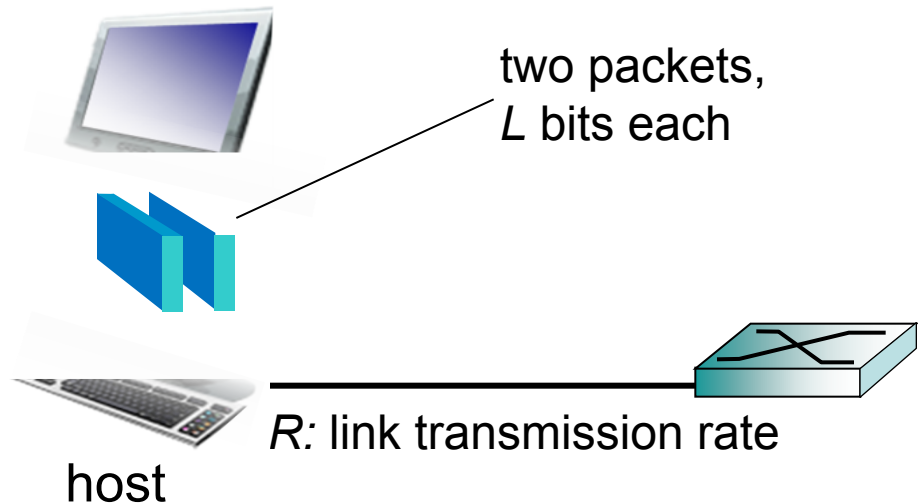
vided by telco (cellular)



Host: sends *packets* of data

host sending function:

- ❖ takes application message
- ❖ breaks into sma
- ❖ transmits packet into access network at *transmission rate* R



- link transmission

rate, aka link

*capacity, aka link
bandwidth*

time needed to
transmit L-bit
packet into link

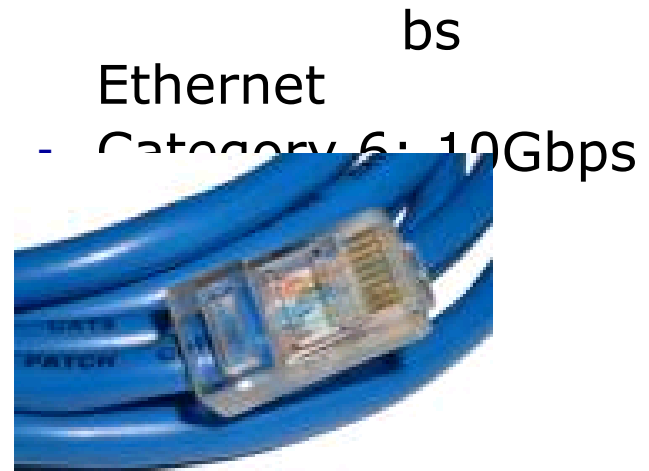
=

$$\frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Physical media

- ❖ **bit**: propagates between transmitter/receiver pairs
- ❖ **physical link**: w
- ❖ **guide**
 - signals propagate in solid media: copper, fiber, coax
- ❖ **unguided media**:
 - signals propagate freely, e.g., radio

twisted pair (TP)
o insulated copper

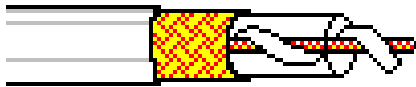


Physical media: coax, fiber

coaxial cable:

- ❖ two concentric copper conductors
- ❖ bidirectional

cabl



fiber optic cable:

- ❖ glass fiber carrying light pulses, each pulse

int
e.g., 10's-
100's Gpbs transmission
rate)

- ❖ low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Physical media: radio

- ❖ signal carried in electromagnetic spectrum
- ❖ no physical “wire”

environment

- reflection
- obstruction by objects
- interference

radio link types:

- ❖ **terrestrial microwave**
 - e.g. up to 45 Mbps channels

- 3G cellular: ~ few Mbps
- ❖ **satellite**
 - Kbps to 45Mbps channel (or multiple smaller channels)
 - 270 msec end-end delay
 - geosynchronous versus low altitude

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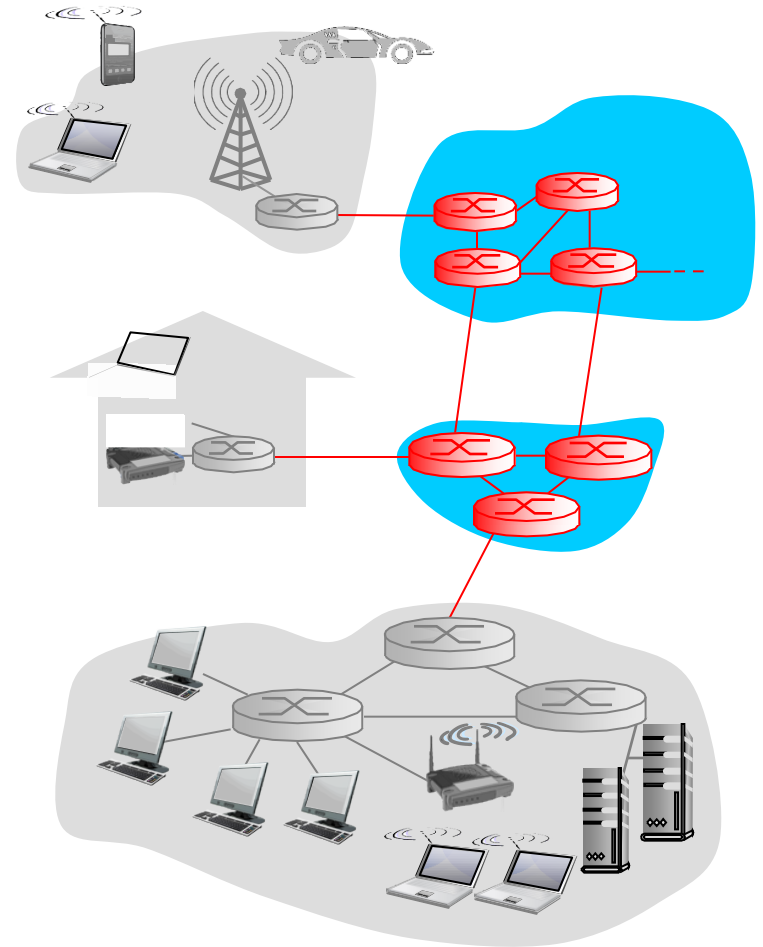
1.7 history

The network core

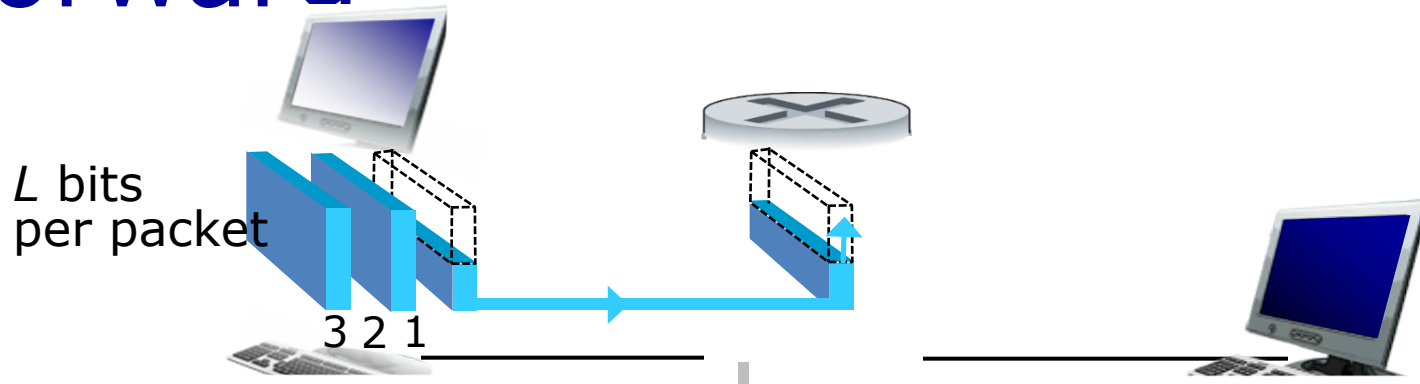
- ❖ mesh of interconnected routers

appli
messages into
packets

- forward packets from one router to the next, across links on path from source to destination
- each packet transmitted at full



Packet-switching: store-and-forward

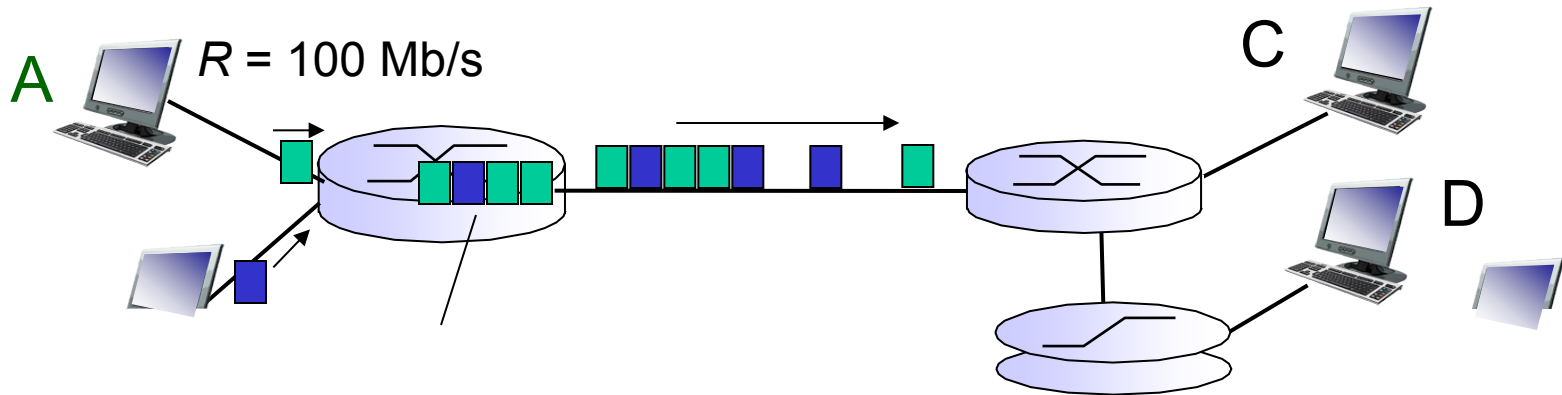


- ❖ takes
transmit (push out) L -
bit packet into link at R
bps
- ❖ *store and forward*:
entire packet must
arrive at router before
it can be transmitted
- ❖ incremental delay = L/R
(assuming zero
propagation delay)

*numerical
example:*

- $L = 7.5$ Mbits
 - $R = 1.5$ Mbps
 - one-hop
transmission delay
= 5 sec
- } more on delay shortly ...

Packet Switching: queueing delay, ~~loss~~



queueing and loss:

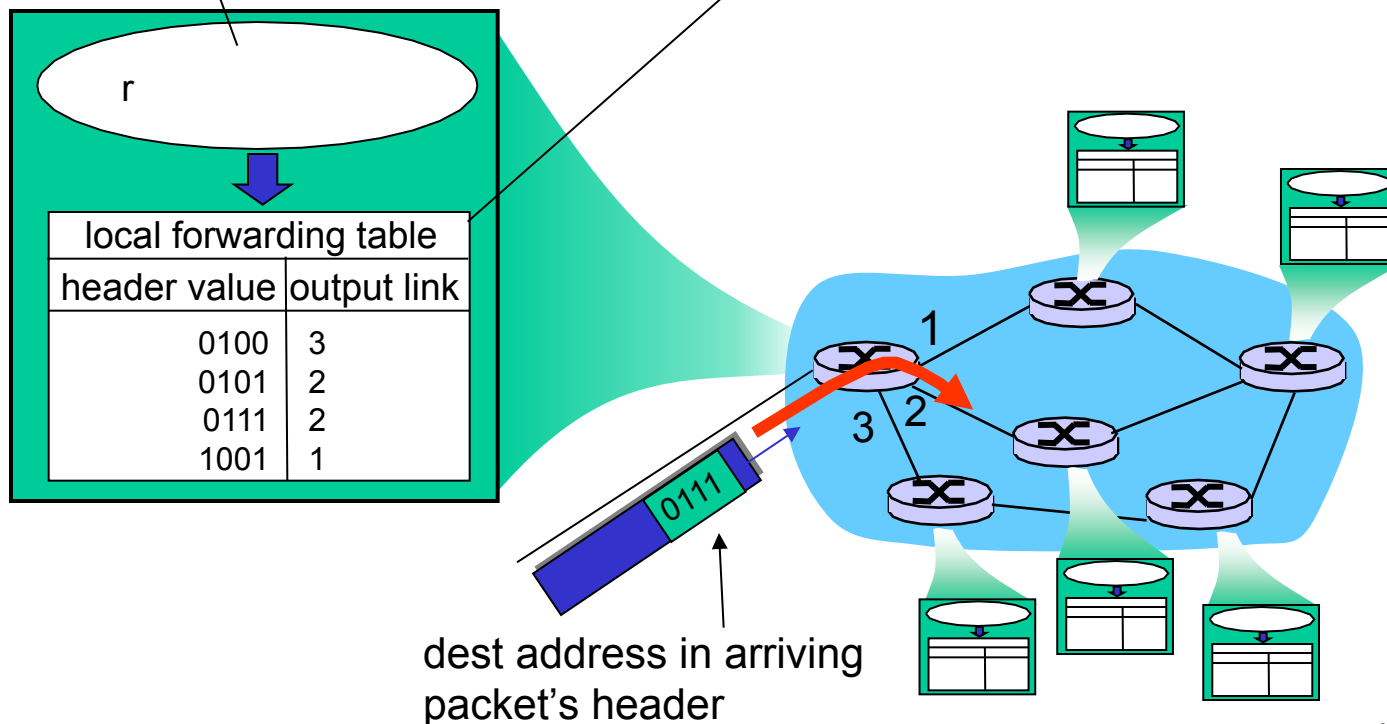
- ❖ If arrival rate (in bits) to link exceeds transmission rate of link for a period of time:
 - packets will queue, wait to be transmitted on link
 - packets can be dropped (lost) if memory (buffer) fills up

Two key network-core functions

routing: determines source-destination route taken by packets

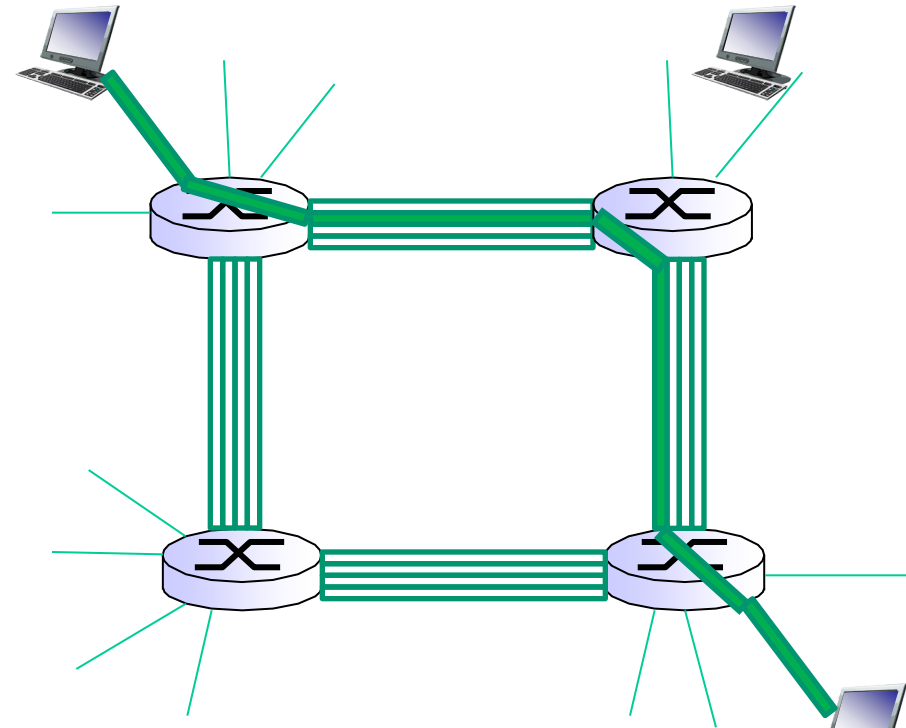
- *routing algorithm*

forwarding: move packets from router's input to appropriate output



Alternative core: circuit switching

end-end resources
allocated to, reserved
for "call" between
source & dest



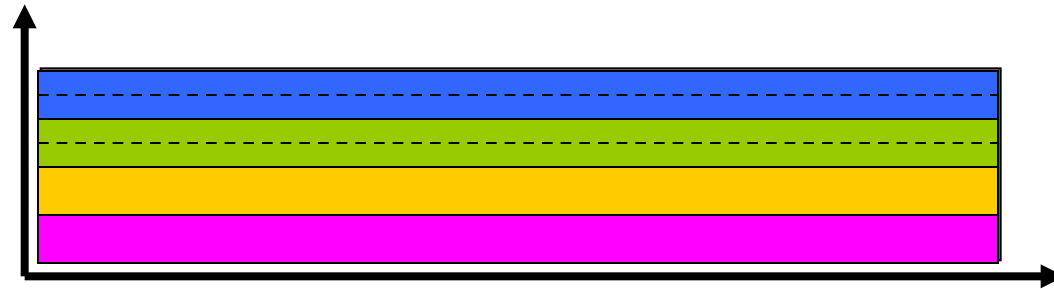
- ❖
 - call
top link and 1st circuit
in right link.
- ❖ dedicated resources: no
sharing
 - circuit-like
(guaranteed)
performance
- ❖ circuit segment idle if not
used by call (*no sharing*)

Circuit switching: FDM versus TDM

FDM

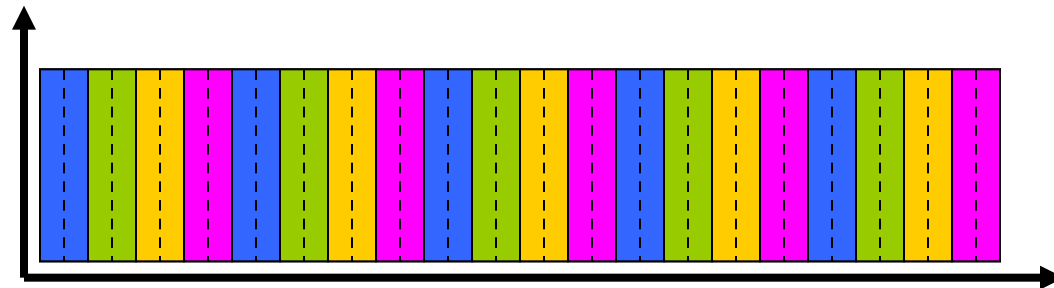
Example:

4 users



TDM

frequency



time

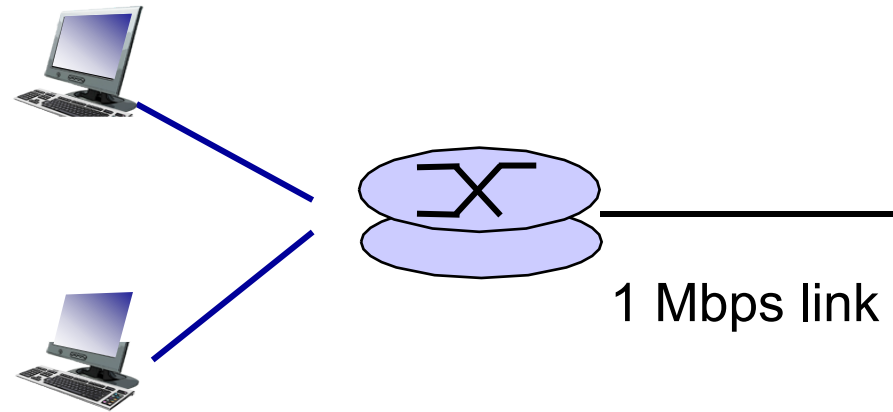
Packet switching versus circuit switching

packet switching allows more users to use network!

example:

- 1 Mb/s link

- activ



❖ *circuit-switching:*

- 10 users

❖ *packet switching:*

- with 35 users,
probability > 10 active
at same time is less
than .0004 *

Q: how did we get value 0.0

Q: what happens if > 35 use

Packet switching versus circuit switching

is packet switching a “slam dunk winner?”

- ❖ great for bursty data
 - resource sharing
 - simpler, no ca



- prot congestion control fer,

❖ Q: How to provide circuit-like behavior?

- bandwidth guarantees needed for audio/video apps

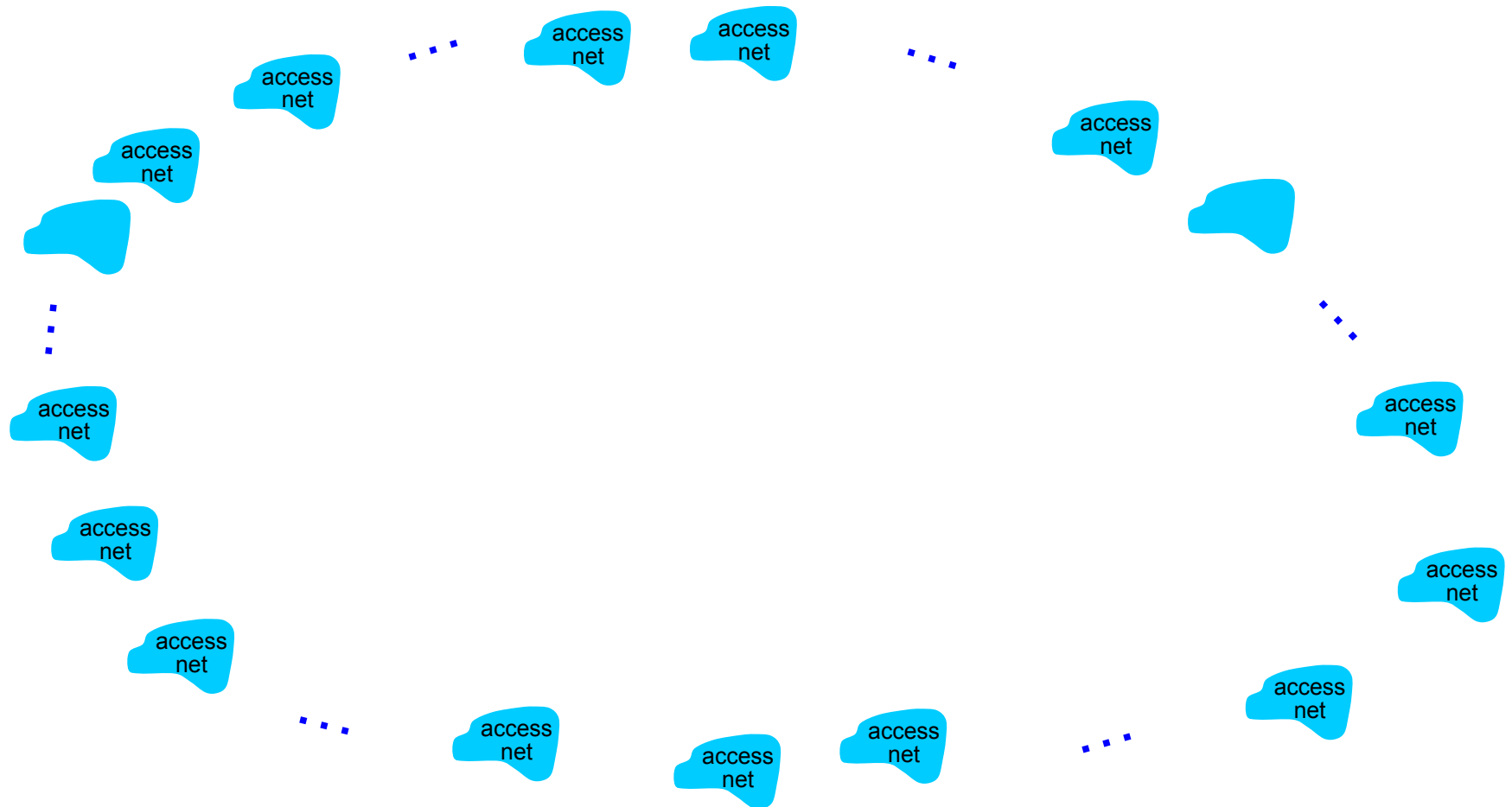
Q: ■ still an unsolved problem (chapter 7)
human analogies of reserved resources (circuit switching) versus on-demand allocation (packet-switching)?

Internet structure: network of networks

- ❖ End systems connect to Internet via **access ISPs** (Internet Service Providers)
 - Residential, company and university ISPs
- ❖ Access ISPs in turn are interconnected.
- ❖ Result is a very complex network
 - ❖ Evolution was driven by **economics** and **national policies**
- ❖ Let's take a stepwise approach to describe current Internet structure

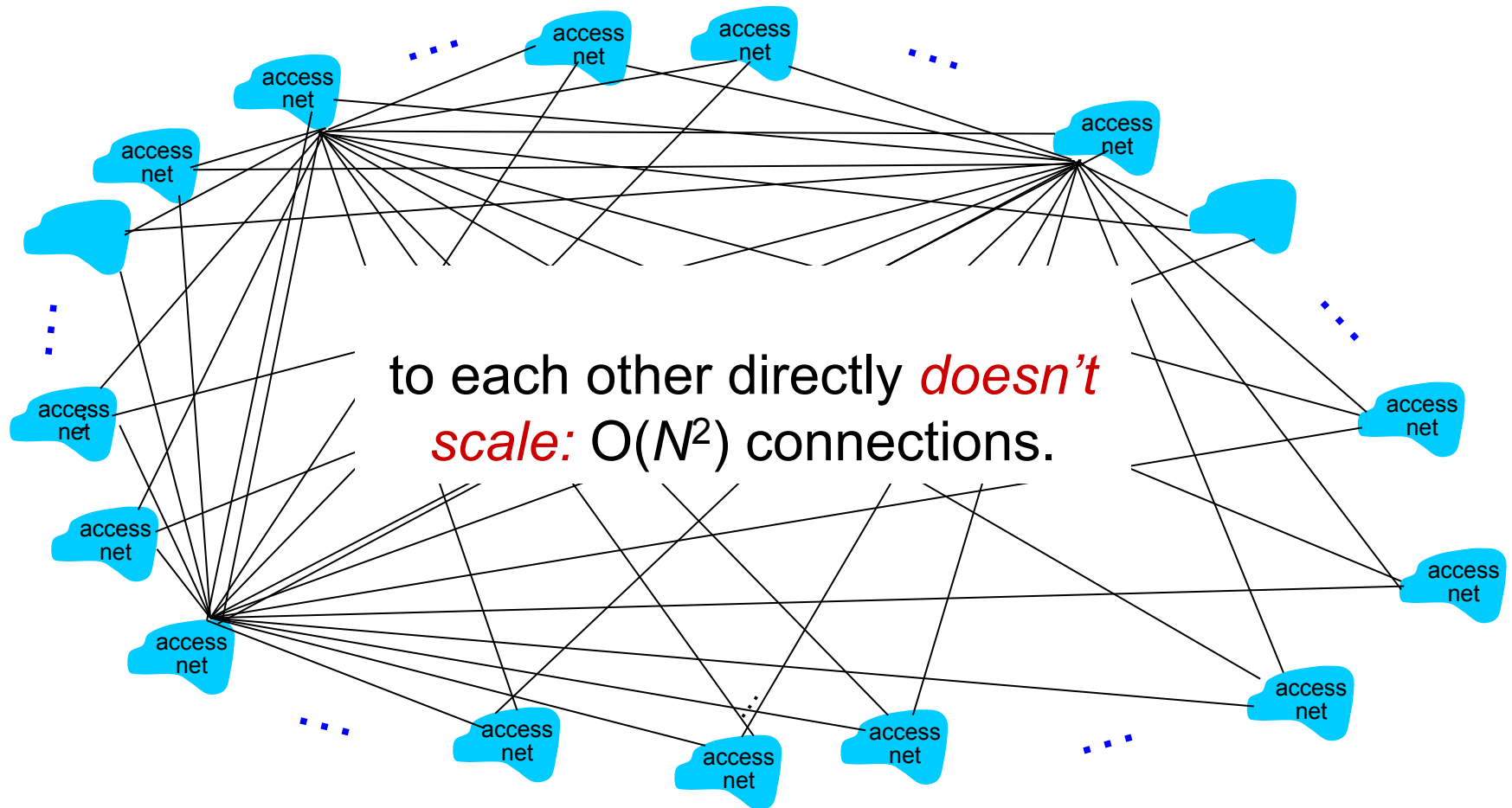
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



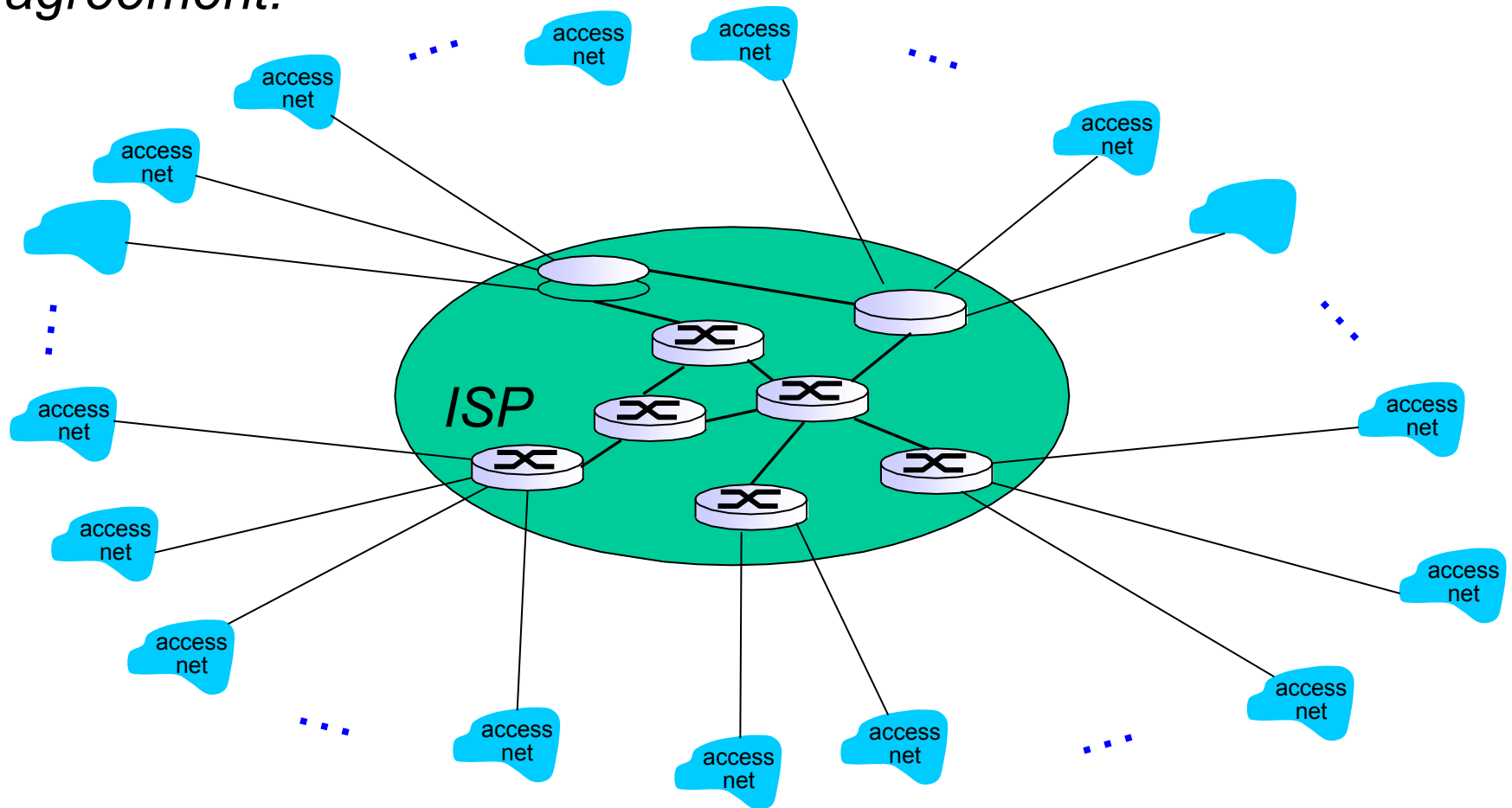
Internet structure: network of networks

Option: connect each access ISP to every other access ISP?



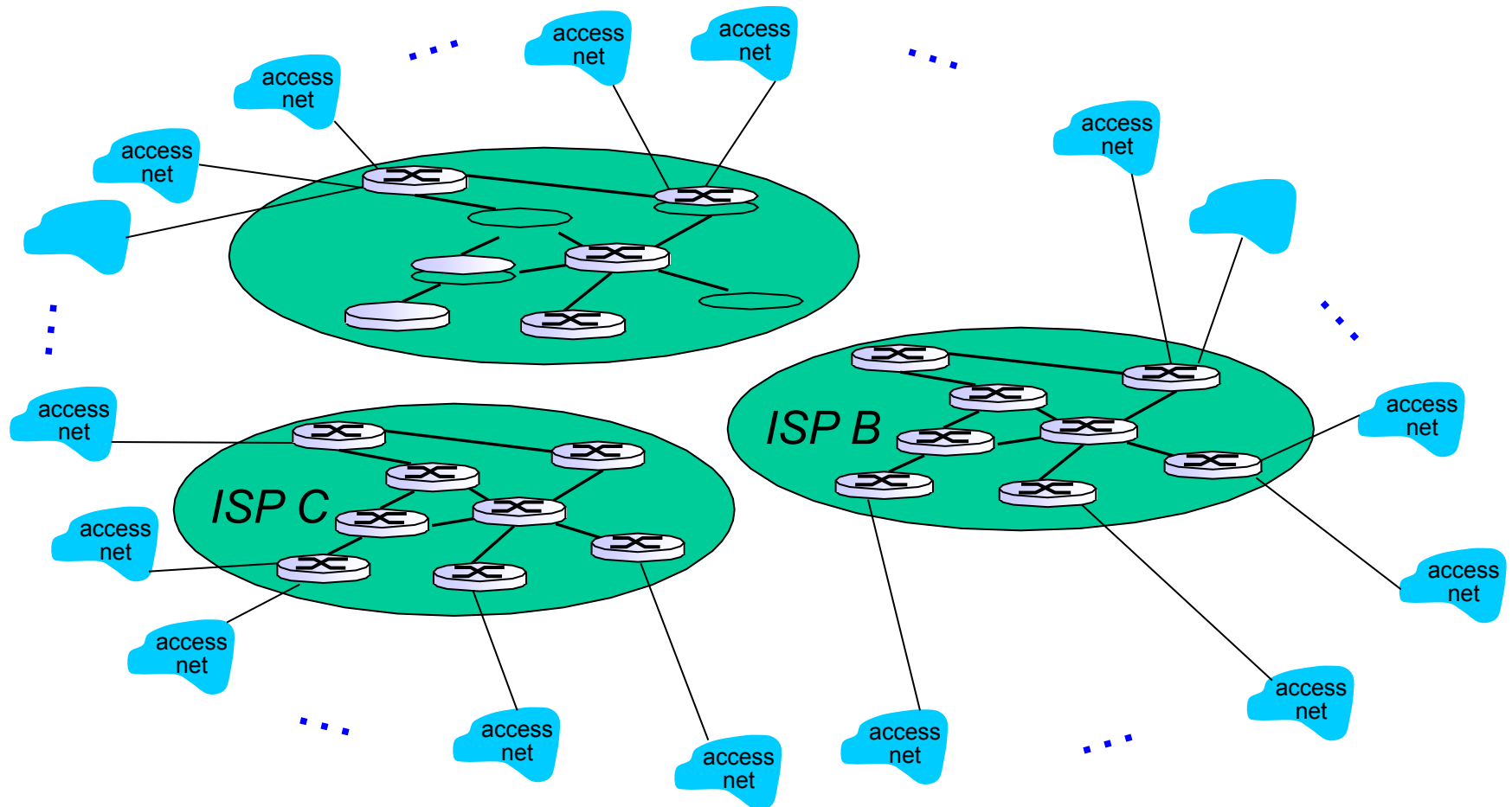
Internet structure: network of networks

Option: connect each access ISP to a global transit ISP? Customer and provider ISPs have economic agreement.



Internet structure: network of networks

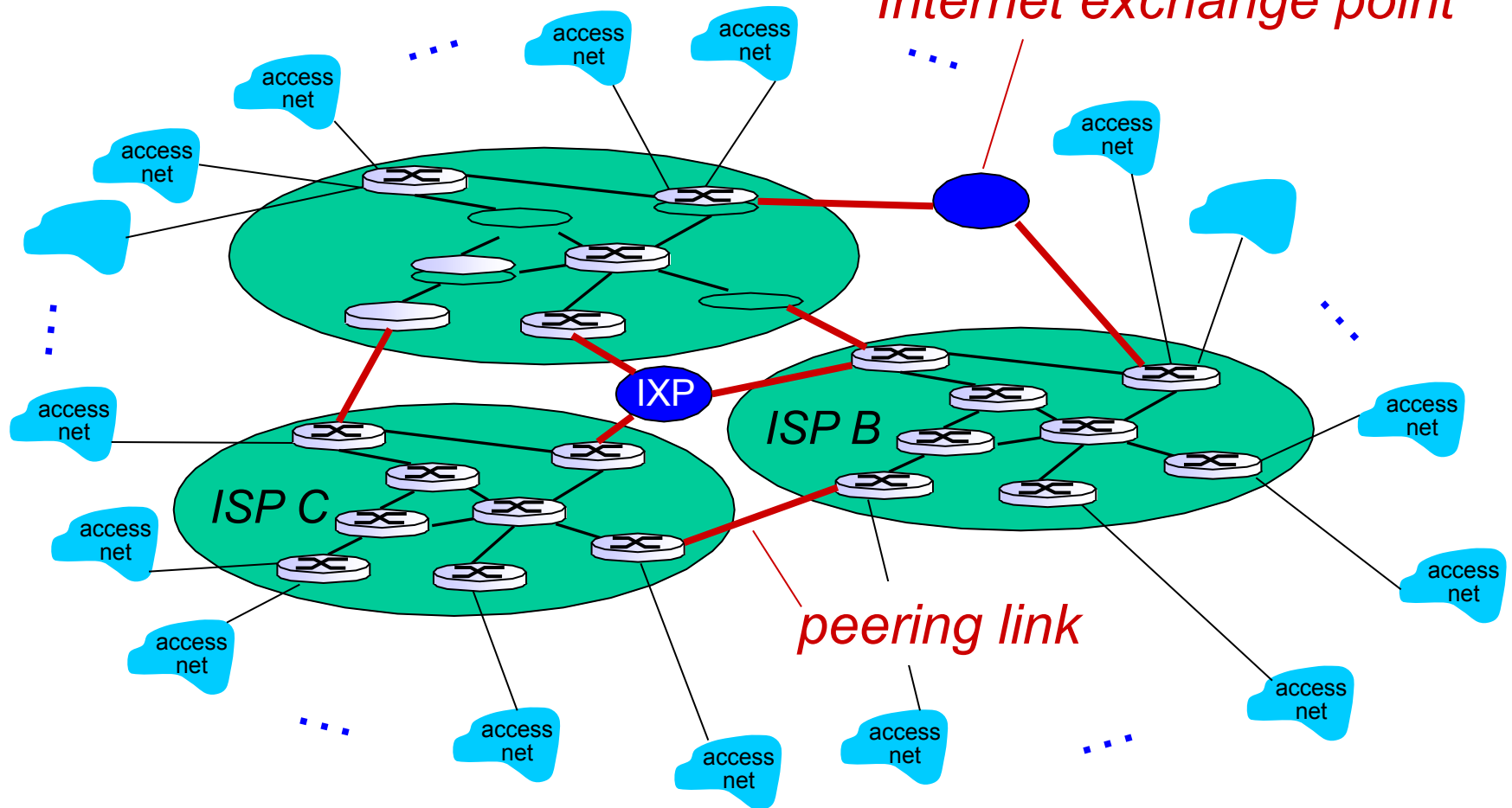
But if one global ISP is viable business, there will be competitors



Internet structure: network of networks

But if one global ISP is viable business, there will be competitors which must be interconnected

Internet exchange point



Internet structure: network of networks

... and regional networks may arise to connect access nets to ISPs

